Transform Yourself

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

A Multi-Agent LLM Framework for Automated Technical Interview Assessment with Real-Time Gaze-Based Proctoring

FIRST REVIEW

PROJECT GUIDE

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Estd : 1984

Objective

- Extend adaptive learning to placement aptitude, coding, and interview rounds.
- Combine live AI interviews with gaze tracking, face auth, phone detection, and tab monitoring.
- Deploy specialized AI agents for resume parsing, dynamic question generation, and feedback.
- Provide teachers with centralized student performance analytics and proctoring logs.

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Project Goal AND ITS SDG

- **Core Goal:** Automate campus placement training using multi-agent AI interviewers and real-time gaze proctoring.
- **Personalized Learning:** Adapt test difficulty and interview questions dynamically based on individual student weak areas.
- **Faculty Efficiency:** Reduce manual mock interview workload for teachers while providing detailed readiness analytics.
- **SDG 4 (Quality Education):**Provide equal access to personalized skill-building tools for all students to master core concepts.
- **SDG 8 (Decent Work & Economic Growth):** Improve student employability and interview confidence to boost campus placement outcomes.

PROBLEM STATEMENT

- **Unscalable Manual Interviews:** Conducting 1-on-1 mock interviews for hundreds of students takes excessive faculty time.
- **Static Assessment Tests:** Standard online tests give identical questions to everyone without addressing individual weak areas.
- **No Live Coding Evaluation:** Multiple-choice quizzes cannot test real code compilation, execution speed, or memory usage.
- **High Malpractice Vulnerability:** Online remote tests lack real-time monitoring for impersonation, phone usage, or tab switching.
- **Lack of Student Analytics:** Teachers lack a unified dashboard to track student readiness metrics before placement drives.

EXISTING SYSTEMS

- **Manual Faculty Mock Interviews:** Relies on professors conducting individual, time-consuming 1-on-1 mock technical and HR interviews.
- **Static Quiz Platforms:**Uses standard online form tools (e.g., Google Forms/Moodle) with fixed, non-adaptive question sets**.**
- **Multiple-Choice Coding Quizzes:** Tests programming knowledge using static option choices rather than live code execution environments.
- **Basic Invigilation & Simple Logging:** Relies on physical room supervision or basic tab-switch counters without computer vision tracking.
- **Isolated Score Reports:** Produces separate marks without providing combined performance analytics or placement readiness scores.

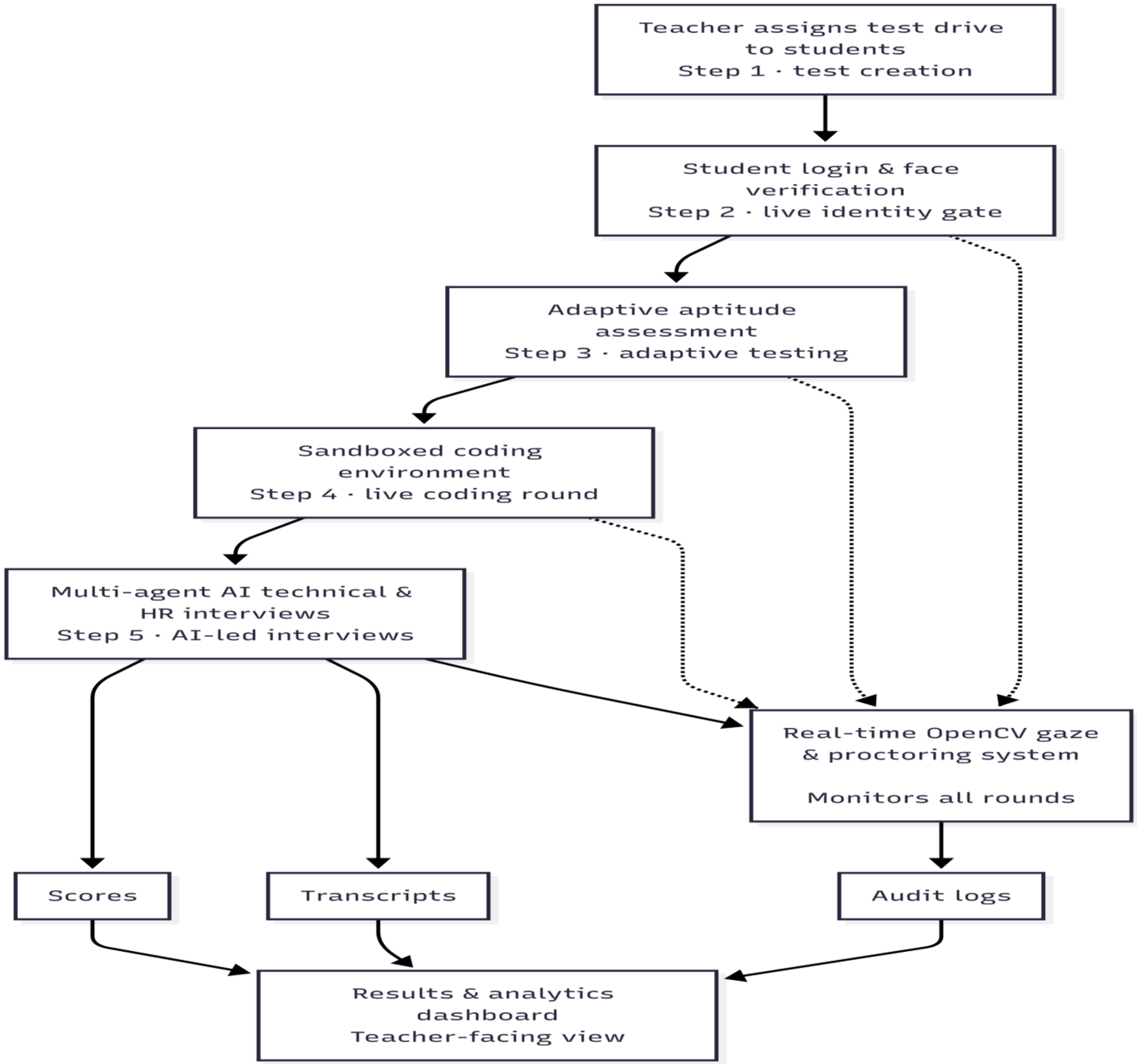
PROPOSED IDEA

- **Integrated 4-Round Assessment Pipeline:** Combines Face Verification, Adaptive Aptitude, Coding Sandbox, Technical, and HR interviews in one platform.
- **Multi-Agent LLM Interviewers:** Uses specialized AI agents for resume parsing, technical stack probing, and STAR behavioral interviews.
- **Adaptive Aptitude Engine:** Automatically shifts question density to target student weak topics based on past performance history.
- **Sandboxed Code Execution Engine:** Provides a Monaco-based editor with real-time compilation, runtime speed, memory, and test case diagnostics.
- **Real-Time Gaze-Based Proctoring:** Monitors eye gaze, head movement, face count, phone presence, and tab switching using CPU-friendly OpenCV vision.

MODULES WITH DESCRIPTION

Modules	Description
1.Biometric Face Verification	Authenticates student identity via webcam capture and 128-d face embedding matching before unlocking test access.
2. Adaptive Aptitude Engine	Analyzes past test attempt history to dynamically generate personalized quiz papers targeting individual weak topics.
3.Sandboxed Code Execution	Compiles multi-language code via Judge0 Docker sandboxes to evaluate runtime speed, memory usage, and test cases.
4. Multi-Agent LLM Interviewer	Parses candidate resumes and orchestrates context-aware technical (DSA/CS Core) and STAR-method behavioral HR interviews.
5.Real-Time Gaze Proctoring	Uses CPU-optimized OpenCV models to track eye gaze deviation, head movement, face count, phone usage, and tab switching.

WorkFlow



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RESEARCH GAP IDENTIFICATION

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Papers (10 /100)	Insights	Results	Research Gap
Journal Article • 10.56294/dm2025981 1. AI-Powered Adaptive Learning Systems in Higher Education: A Scoping Review of Implementation and Impact on Academic Performance Iván Claudio Suazo Galdames, Alain Manuel Chaple Gil, Iván Claudio Suazo Galdames +1 more 22 Oct 2025 - Data & metadata PDF Summary Podcast Chat 66	The review systematically analyzes 14 studies on AI-powered adaptive learning platforms, highlighting their positive impacts on academic performance, engagement, and satisfaction across various disciplines, while addressing implementation challenges and the need for equitable strategies in higher education.	• The implementation of AI-powered adaptive learning systems in higher education consistently improved academic performance across various disciplines. • Students using adaptive learning platforms demonstrated significant gains in engagement, retention, and satisfaction compared to traditional learning environments. • Specific studies reported statistically significant higher post-test scores and improved self-efficacy among students utilizing adaptive learning technologies. • Barriers to implementation included faculty resistance, ethical concerns regarding data privacy and algorithmic bias, and regional disparities in adoption, particularly in Latin America and Sub-Saharan Africa. • The review highlighted the need for inclusive design, long-term evaluation, and equitable implementation strategies to maximize the benefits of adaptive learning systems. Save to Notebook	• The study focuses on AI-powered adaptive learning for academic education but does not explore its application in placement preparation, including adaptive aptitude tests, technical interviews, coding assessments, and HR interviews. • Existing adaptive learning systems primarily personalize learning content but lack real-time AI-based interview evaluation and intelligent proctoring, such as gaze tracking, face verification, phone detection, and cheating prevention during online assessments. • Ethical concerns such as data privacy and algorithmic bias remain unresolved, indicating a need for further exploration of these issues in the context of adaptive learning systems. • Faculty resistance to adopting these systems presents a barrier that has not been adequately addressed, suggesting a need for strategies to overcome this challenge. • The review highlights the necessity for long-term evaluation and equitable implementation strategies, which remain underexplored in the current literature on adaptive learning systems.

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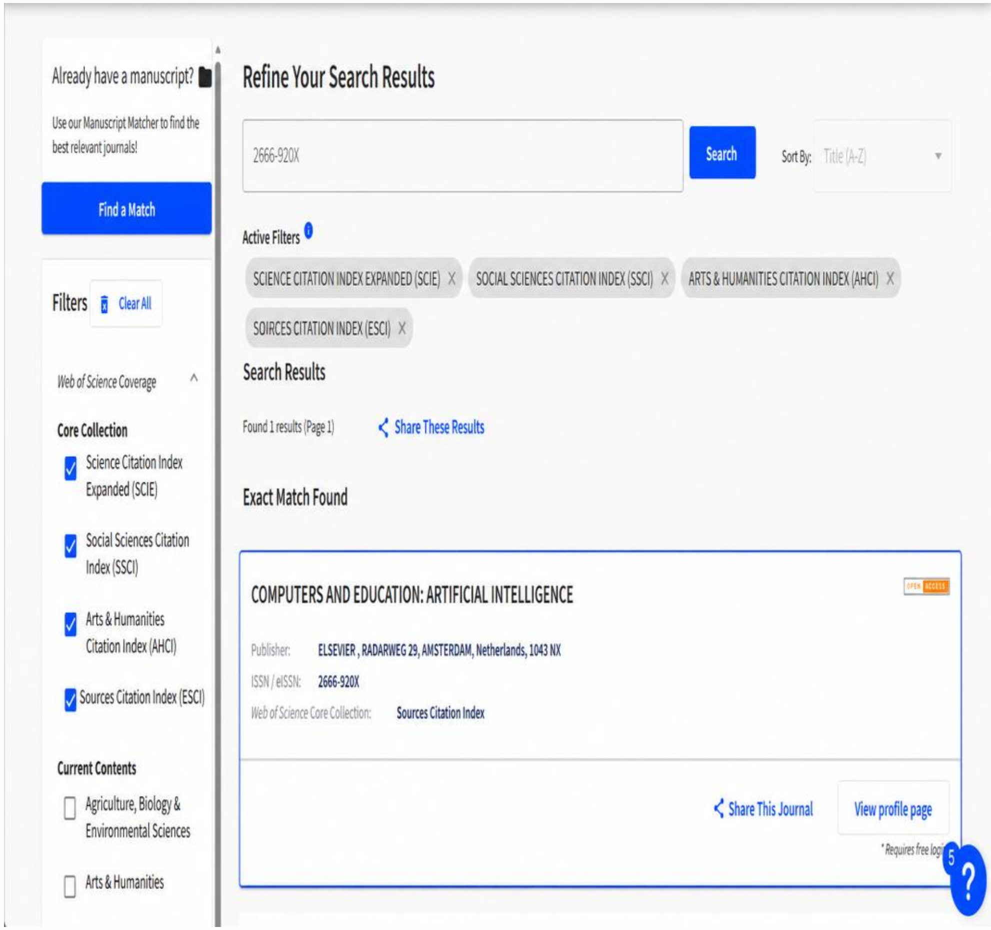
SCIE PAPER & CRITIQUE



Innovation:	8.8/10
Research Novelty:	8.2/10
Problem Selection:	9.6/10
Solution Design:	9.0/10
AI Architecture:	8.7/10
System Infrastructure & Routing:	8.4/10
Vector / RAG Implementation:	7.5/10
Technology Stack:	9.0/10
Technical Complexity:	8.9/10
Software Engineering Quality:	8.5/10
Scalability:	8.0/10
Security & Safety:	8.2/10
Explainability:	9.0/10
Practical Usability:	9.2/10
Publication Potential:	8.5/10
Overall Project Quality:	8.8/10

SCIE PAPER

CRITIQUE RATING

FUTURE SCOPE

- **Voice & Tone Analysis:** Evaluate speech confidence, pitch, and clarity during AI interviews.

- **Architecture Diagram Grading:** Grade digital or hand-drawn system design diagrams using Vision-LLMs.

- **University LMS Integration:**Connect seamlessly with Moodle and Canvas for automated score syncing.

- **AI Code Optimization Hints:** Provide real-time feedback on code complexity, memory, and refactoring.

- **Multi-Lingual Support:**Conduct AI technical and HR interviews in regional languages.

THANK YOU